WORKED SOLUTION

STEP I 2015, Question 12

Probability and Statistics • Thinning a Poisson Distribution

STEP I 2015, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q12

The key idea

Conditioning on the number of arrivals turns surgery into a binomial count. Summing over all possible arrival numbers (the law of total probability) then gives the distribution of the number needing surgery, and the sum collapses into an exponential series. For the final part, the key fact is that if two independent Poisson counts with the same mean are conditioned on their total, the split between them is binomial.

Solution

Let X be the number of casualties arriving on a given day, and let Y be the number of them who need surgery.

Part (i)

Given $X = n$, each of the n casualties needs surgery with probability $\frac{1}{4}$, independently of the others. So, conditionally, $Y \sim \mathrm{B}(n, \frac{1}{4})$, and for $0 \le r \le n$

$$\mathrm{P}(Y = r \mid X = n) = \binom{n}{r}\left(\frac{1}{4}\right)^r\left(\frac{3}{4}\right)^{n-r} = \binom{n}{r}\frac{3^{n-r}}{4^n}.$$

(The probability is 0 if $r > n$.)

Part (ii)

Fix $r \ge 0$. Since $Y = r$ can only happen when $X \ge r$, the law of total probability gives

$$\begin{aligned}\mathrm{P}(Y = r) &= \sum_{n=r}^{\infty} \mathrm{P}(X = n)\,\mathrm{P}(Y = r \mid X = n)\\ &= \sum_{n=r}^{\infty} \frac{\mathrm{e}^{-8}8^n}{n!} \cdot \frac{n!}{r!\,(n-r)!}\left(\frac{1}{4}\right)^r\left(\frac{3}{4}\right)^{n-r}.\end{aligned}$$

The $n!$ cancels. Now split $8^n = 8^r \cdot 8^{n-r}$, so that $8^r(\frac{1}{4})^r = 2^r$ and $8^{n-r}(\frac{3}{4})^{n-r} = 6^{n-r}$:

$$\begin{aligned} \mathrm{P}(Y = r) &= \frac{\mathrm{e}^{-8}\,2^r}{r!} \sum_{n=r}^{\infty} \frac{6^{n-r}}{(n-r)!} \\ &= \frac{\mathrm{e}^{-8}\,2^r}{r!} \sum_{k=0}^{\infty} \frac{6^k}{k!} \qquad (k = n-r) \\ &= \frac{\mathrm{e}^{-8}\,2^r}{r!}\,\mathrm{e}^6 = \frac{\mathrm{e}^{-2}\,2^r}{r!}. \end{aligned}$$

This holds for every $r = 0, 1, 2, \ldots$, so the number needing surgery each day follows a Poisson distribution with mean $8 \times \frac{1}{4} = 2$.

Ask yourself Where did the e^6 come from?

Once everything that depends only on r is taken outside the sum, what is left is $\sum 6^k/k!$, the exponential series for e^6. Splitting 8^n as $8^r \cdot 8^{n-r}$ is exactly what makes the remaining sum depend on $n - r$ alone.

Part (iii)

Let M and T be the numbers needing surgery on Monday and Tuesday. By part (ii) each is $\mathrm{Po}(2)$. They are independent, since the arrivals on different days are independent and casualties need surgery independently of one another. Then

$$\mathrm{P}(M = 8 \mid M + T = 12) = \frac{\mathrm{P}(M = 8,\ T = 4)}{\mathrm{P}(M + T = 12)}.$$

By independence the numerator is

$$\mathrm{P}(M = 8)\,\mathrm{P}(T = 4) = \frac{\mathrm{e}^{-2}2^8}{8!} \cdot \frac{\mathrm{e}^{-2}2^4}{4!} = \frac{\mathrm{e}^{-4}\,2^{12}}{8!\,4!}.$$

For the denominator, the sum of independent Poisson variables is Poisson, with the means added, so $M + T \sim \mathrm{Po}(4)$. If you would rather not quote this, it follows directly:

$$\mathrm{P}(M + T = 12) = \sum_{m=0}^{12} \frac{\mathrm{e}^{-2}2^m}{m!} \cdot \frac{\mathrm{e}^{-2}2^{12-m}}{(12-m)!} = \frac{\mathrm{e}^{-4}2^{12}}{12!} \sum_{m=0}^{12} \binom{12}{m} = \frac{\mathrm{e}^{-4}\,4^{12}}{12!},$$

using $\sum_m \binom{12}{m} = 2^{12}$. Dividing,

$$\begin{aligned} \mathrm{P}(M = 8 \mid M + T = 12) &= \frac{\mathrm{e}^{-4}\,2^{12}}{8!\,4!} \cdot \frac{12!}{\mathrm{e}^{-4}\,4^{12}} \\ &= \binom{12}{8}\left(\frac{1}{2}\right)^{12} = \frac{495}{4096}. \end{aligned}$$

Since $495 = 3^2 \cdot 5 \cdot 11$ is odd and $4096 = 2^{12}$, the fraction is in lowest terms. So the required probability is $\dfrac{495}{4096}$.

Ask yourself Why does the answer look like a binomial probability?

Given the total of 12, each of those 12 surgical cases is equally likely to have come from Monday or Tuesday, because the two days have the same mean. So the Monday count is $\mathrm{B}(12, \frac{1}{2})$. Notice that the original mean of 8 has disappeared completely.

Checks

- The mean in part (ii) agrees with $\mathrm{E}(Y) = \mathrm{E}\big(\mathrm{E}(Y \mid X)\big) = \mathrm{E}(X/4) = 2$.
- Summing the part (ii) series numerically for $r = 0, \ldots, 14$ reproduces $\mathrm{e}^{-2}2^r/r!$ to within rounding error. For example, $\mathrm{P}(Y = 0) = \sum_n \mathrm{e}^{-8}8^n(\frac{3}{4})^n/n! = \mathrm{e}^{-8}\mathrm{e}^{6}$.
- The conditional probabilities $\binom{12}{m}/2^{12}$ for $m = 0, \ldots, 12$ sum to 1, and $495/4096 \approx 0.121$. A computer simulation of the full model (Poisson arrivals, then surgery with probability $\frac{1}{4}$ each) gave a value consistent with this.

Common slips

- Starting the sum in part (ii) at $n = 0$ instead of $n = r$: there is no way for r casualties to need surgery if fewer than r arrive.
- Using $\mathrm{Po}(8)$ for each day in part (iii). The conditioning is on surgical cases, so the relevant mean is 2. (Here the means cancel anyway, but the working should be right.)
- Leaving the answer as $\binom{12}{8}/2^{12}$ or $1980/16384$ when lowest terms were asked for.

Beyond STEP

Part (ii) is called *thinning*: if each point of a Poisson process is kept independently with probability p, the points kept form a Poisson process with rate multiplied by p, and, perhaps surprisingly, it is independent of the points discarded. Part (iii) is the reverse of this: conditioned on their total, independent Poisson counts split multinomially. These two facts underpin the theory of Poisson processes, which you meet in a first probability course. Grimmett and Welsh, *Probability: An Introduction*, covers both.

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